Math 261B: Project Proposal - AI Sentiments

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Research Question

The research question of interest is "Is there an evaluative bias towards AI-generated artwork? How do the influences of perceived effort, perceived threat, and emotional engagement contribute to this bias?"

Identification of Related Literature

The main article of interest is "Will people embrace AI art? Deconstructing psychological barriers in human appraisal of AI-labeled artworks" [1] by Zhang et al. This paper investigates the psychology behind the bias against AI-generated artworks. There are five studies conducted in this paper, but our focus will be on the first four studies. Study 1 confirms the existence of bias towards AI-generated work. Studies 2, 3, and 4 examine how perceived effort of artworks, perceived threat, and emotional engagement are respectively influenced by the AI label. The results showed that they believe that AI puts in less effort, they are threatened by AI's autonomy, and they believe AI lacks emotional capabilities. The data for this paper is publicly available and will be the basis for the rest of the paper.

"Artificial Intelligence, Artists, and Art: Attitudes Toward Artwork Produced by Humans vs. Artificial Intelligence" [2] studies how AI-art is perceived overall and how the knowledge of the artist's identity affects the evaluation of a piece.

"AI-generated vs. Human Artworks" [3] defines a different set of responses as a basis for content evaluation: liking, perceived beauty, novelty, and meaning.

Study Design

The goal of the study was to examine if there exists inherent negative bias towards AI-generated artwork. Based on 4 formal experiments, three of which were preregistered, those with favorable attitudes towards AI showed less bias (Study 1), perceived effort was positively correlated with favorability (Study 2), perceived threat by AI increased bias (Study 3), and people who paid less to emotional aspects of AI paintings led to greater bias (Study 4).

Study 1 recruited 60 subjects and showed them 12 statements. Participants were required to rate each statement on a 1-5 scale (1 = strongly disagree, 5 = strongly disagree).

Study 2 had 60 participants who were required to rate 78 paintings based on creativity, aesthetic, liking, and delicacy on a 1-5 scale like Study 1. They were also asked to rank the perceived effort put into the artworks. The paintings were both AI-generated and human-made and were randomized in order. They were told that the model usually takes 30-40 minutes to generate a painting.

Study 3 recruited 120 participants, who were then randomly divided into two groups: a control group and a threat group. Participants in the threat group were given a prompt that contained a statement that threatened human employment, while the control group had no such statement. After that, both groups were required to choose in 40 rounds, between 2 labeled paintings (1 human, 1 AI, 80 paintings total), which one was more creative, aesthetic, delicate, and which one did the participant prefer.

Study 4 recruited 85 participants and divided them into two groups. One group was asked to pay more attention to the story and emotions of the paintings, while the other group was asked to focus more on the techniques and details. Then, the groups were asked to evaluate 30 labeled paintings, half of which were generated by AI. They were asked to evaluate the paintings on a scale from 1-7 based on aesthetics, profundity, creativity, aesthetic emotion dimensions (liking, awe, enchantment, joy, vitality, interest, insight, uneasiness, boredom, and sadness), emotional perception (valence and arousal), and their focus on the painting.

Analysis Plan

The data is made publicly available from the first paper by Zhang et al. We will be reanalyzing their work done with ANOVA and a regression model including main and interaction effects. The participant judging the images will be a random effect, and the source of the image will be a fixed effect. We will also assess the t-tests on the differences in ratings (human vs. AI-label). The authors returned multiple post-hoc confidence intervals comparing the differences in creativity, aesthetics, liking, and delicacy (human minus AI); it is unclear if they performed multiple testing, so we will attempt to reconstruct the intervals using Tukey's Honest Significant Difference and the Bonferroni correction.

Our selected article includes a power analysis for the sample size they chose in all studies. Authors used a statistical program called G-power to find the appropriate sample size for an effect size of 0.37 at 80% power. It is unclear how they used past literature to estimate the effect size of this experimental setup. Likewise for studies 2-4, there are estimated effect sizes based on previous research that the authors used as a basis for the sensitivity power analysis.

References

[1] Wenyu Zhang, Ling Huang, Jiaxin Ding, Cong Xie, Jiaxin Mu, Ning Hao, “Will people embrace AI art? Deconstructing psychological barriers in human appraisal of AI-labeled artworks,” *Computers in Human Behavior Reports*, vol 22, May 2026, doi: https://doi.org/10.1016/j.chbr.2026.101023.

[2] I. Rae, “The Effects of Perceived AI Use On Content Perceptions,” in *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, in CHI ’24. New York, NY, USA: Association for Computing Machinery, May 2024, pp. 1–14. doi: 10.1145/3613904.3642076.

[3] M. Ragot, N. Martin, and S. Cojean, “AI-generated vs. Human Artworks. A Perception Bias Towards Artificial Intelligence?,” in *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, in CHI EA ’20. New York, NY, USA: Association for Computing Machinery, Apr. 2020, pp. 1–10. doi: 10.1145/3334480.3382892.